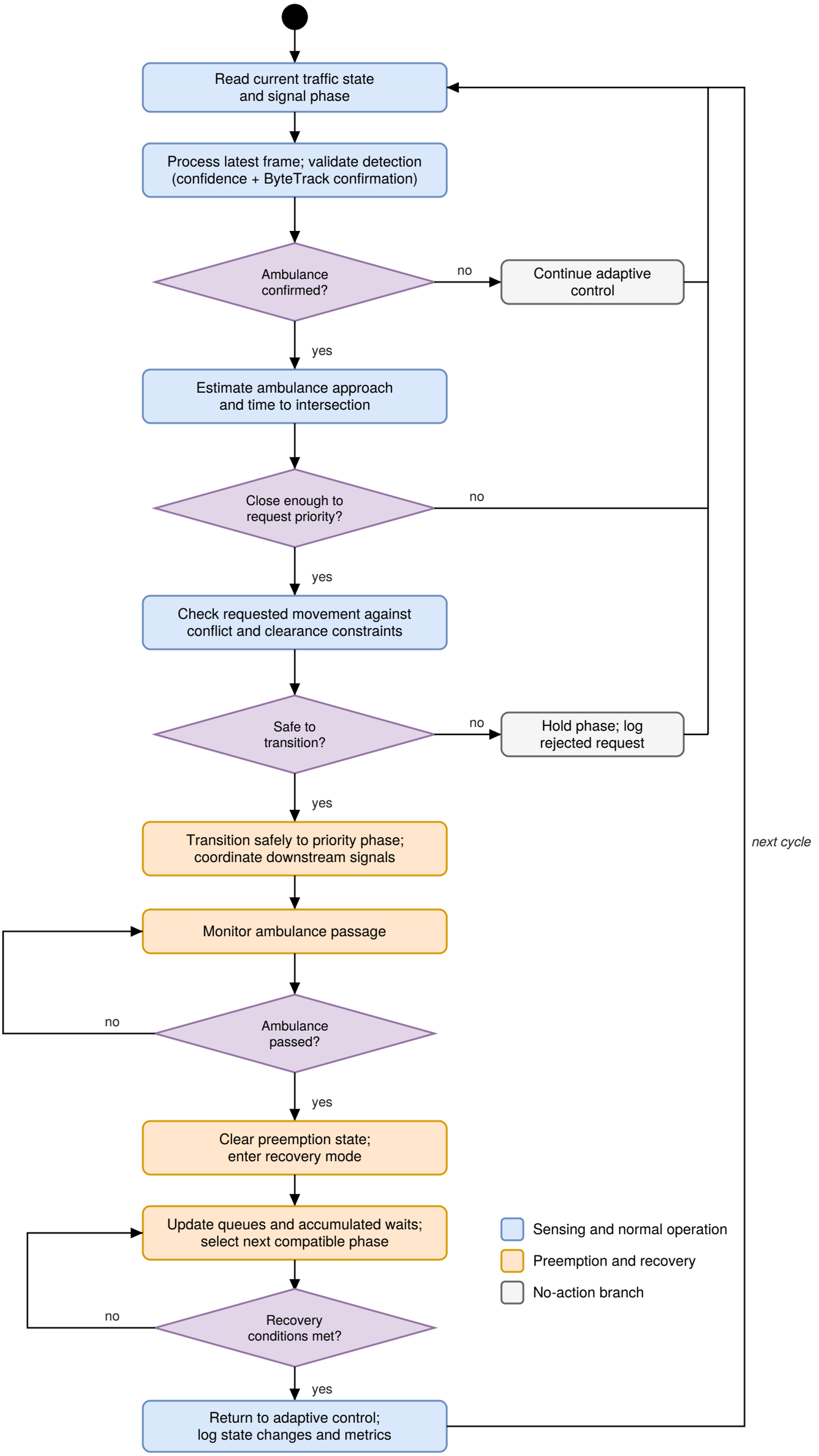
Read current traffic state and signal phase
Process latest frame; validate detection (confidence + ByteTrack confirmation)
Ambulance confirmed?
no
Continue adaptive control
yes
Estimate ambulance approach and time to intersection
Close enough to request priority?
no
yes
Check requested movement against conflict and clearance constraints
Safe to transition?
no
Hold phase; log rejected request
yes
Transition safely to priority phase; coordinate downstream signals
Monitor ambulance passage
no
Ambulance passed?
yes
Clear preemption state; enter recovery mode
Update queues and accumulated waits; select next compatible phase
no
Recovery conditions met?
yes
Return to adaptive control; log state changes and metrics
next cycle
Sensing and normal operation
Preemption and recovery
No-action branch